**a** **BFF contracts: detection and persistence**

| Variant | Worlds N | Modal worlds, n | Population worlds, n | Collapse n / pop. n |
|---|---|---|---|---|
| Standard | 24 | 9 | 7 | 0/7 |
| Literal, no wrap | 12 | 12 | 5 | 5/5 |
| Standard, harmless | 12 | 7 | 3 | 0/3 |
| Pointer wrap | 24 | 19 | 19 | 0/19 |
| Wrap + literal | 12 | 12 | 12 | 12/12 |
| Wrap + literal, harmless | 12 | 12 | 12 | 0/12 |

Harmless: unmatched brackets do not halt; literal emits operands

BFF = Brainfuck-family machine; wrap permits pointer wrap

Modal: generation-2 parent-byte-match gain ≥0.3 (headroom-normalized)
Population: ≥50% cells heritable (same gain ≥0.3 criterion)
Later collapse: terminal heritable-cell fraction <10% after population detection

**b** **First-modal tapes: fetch / inflow counts**

| Fetch/256 | H (bits) | Tapes, n |
|---|---|---|
| **Standard (n=9)** | | |
| 0.00000 | 0.00000 | 4 |
| 0.00000 | 7.24969 | 1 |
| 0.00391 | 0.03687 | 2 |
| 0.00391 | 2.06372 | 1 |
| 0.19531 | 7.39789 | 1 |
| **Literal, no wrap (n=12)** | | |
| 1.00000 | 7.97656 | 1 |
| 1.00000 | 7.98143 | 2 |
| 1.00000 | 7.98438 | 2 |
| 1.00000 | 7.99219 | 2 |
| 1.00000 | 8.00000 | 5 |
| **Wrap + literal, harmless (n=12)** | | |
| 1.00000 | 0.00000 | 4 |
| 1.00000 | 0.03687 | 6 |
| 1.00000 | 0.07373 | 1 |
| 1.00000 | 0.11056 | 1 |

| Fetch/256 | H (bits) | Tapes, n |
|---|---|---|
| **Pointer wrap (n=19)** | | |
| 0.00000 | 0.00000 | 11 |
| 0.00000 | 0.03687 | 2 |
| 0.00000 | 1.92163 | 1 |
| 0.00000 | 2.52233 | 1 |
| 0.00000 | 7.07753 | 1 |
| 0.00000 | 7.16823 | 1 |
| 0.00781 | 0.07373 | 1 |
| 0.00781 | 1.73558 | 1 |
| **Wrap + literal (n=12)** | | |
| 1.00000 | 0.58729 | 3 |
| 1.00000 | 0.62380 | 1 |
| 1.00000 | 0.69675 | 1 |
| 1.00000 | 0.73319 | 1 |
| 1.00000 | 0.76960 | 2 |
| 1.00000 | 0.80599 | 2 |
| 1.00000 | 0.84236 | 1 |
| 1.00000 | 0.98759 | 1 |

64 first-modal tapes, one/world; n counts each fetch/H pair
Fetch: encounters with any partner fetch /256; values rounded to 5 decimals
H = H(offspring | fixed parent), bits; 256 uniform partners
Five variants shown; Standard, harmless has no entropy archive

**c** **Open fetch; zero inflow for r ≥ 2**

| Operand copies r | Worlds | H bits | Fetch/256 |
|---|---|---|---|
| 1 | 12 | 8.0 | 1.00 |
| 2 | 6 | 0.0 | 1.00 |
| 3 | 6 | 0.0 | 1.00 |

Literal, no wrap; r = 1 baseline; r counts operand copies

Literal P a b emits a,b repeated r times

World medians: conditional H and any-partner-fetch encounters/256
r≥2: zero partner inflow with fetch in every encounter

**d** **Halting dial: all worlds included in counts**

| Halt p | Worlds N | P persistence worlds, n/N | Median final P share all worlds |
|---|---|---|---|
| 0 | 12 | 12/12 | 0.984 |
| 0.01 | 6 | 6/6 | 0.984 |
| 0.03 | 6 | 6/6 | 0.984 |
| 0.1 | 6 | 6/6 | 0.984 |
| 0.3 | 6 | 5/6 | 0.102 |
| 1 | 12 | 0/12 | 0.002 |

Background: Wrap + literal; p = 0 harmless, p = 1 halting

p = probability an unmatched bracket halts; P = literal-copy instruction
Persistence: P-tape share ≥10% at 16,384 BFF population updates (epochs)

**e** **First / final inflow and fetch**

| Variant | Worlds n | H bits first / final | Fetch/256 first / final |
|---|---|---|---|
| Literal, no wrap | 12 | 7.99 / 7.99 | 1.00 / 1.00 |
| Standard | 9 | 0.04 / 0.00 | 0.00 / 0.00 |
| Pointer wrap | 19 | 0.00 / 0.00 | 0.00 / 0.00 |
| Wrap + literal | 12 | 0.75 / 0.64 | 1.00 / 1.00 |
| Wrap + literal, harmless | 12 | 0.04 / 0.00 | 1.00 / 1.00 |

First = first-modal tape; final = terminal modal tape

Five variants shown; Standard, harmless has no entropy archive
Paired world medians; fetch = any-partner-fetch encounters /256

**f** **Registered tests of partner inflow (H)**

| Per-world criterion H in bits | Passing worlds, n/N | Support / reject % passing | Verdict |
|---|---|---|---|
| Q1 Z80 first H>1 | 80/80 | ≥90% / ≤60% | met |
| Q2 Z80 loop final H<0.5<br>67 worlds: loop final tapes | 63/67 | ≥90% / ≤60% | met |
| Q3 Z80 H_first − H_final ≥2 | 63/80 | ≥75% / ≤50% | met |
| Q4a BFF standard first H<0.5 | 6/9 | ≥90% / ≤60% | between |
| Q4b BFF pooled literal variants: first H>1 | 12/36 | ≥90% / ≤60% | rejected |

Q4b: Literal, no wrap; Wrap + literal; Wrap + literal, harmless
12 worlds each; 36 total; same names as a, b and e

Q1–3: Z80 Stage G (80 worlds); Q2: 67 worlds with loop-bearing final tapes
Q4a: Standard BFF; Q4b: the three pooled variants named above
Support/reject use pass/world fraction; between = neither threshold